Aadhaar Data Analysis: Project Explained

This document explains the project from the problem to the final decision. It explains each concept, the reason for its use, and its effect on the result.

Each chapter has this structure:

1. The problem
2. The concept
3. A worked example
4. The use in this project
5. What can go wrong
6. Questions that you can now answer

Part A: The problem

1. Problem statement

Title: Unlocking Societal Trends in Aadhaar Enrolment and Updates.

Task: Identify meaningful patterns, trends, anomalies or predictive indicators. Translate them into clear insights or solution frameworks that can support decisions and system improvements.

The real decision: UIDAI has limited staff, equipment and budget. UIDAI must decide where to use them.

The project answers four questions:

No.	Question	Method
1	Can we trust the data?	Data audit and cleaning
2	Which areas need action?	Metrics and analysis
3	Which districts and dates are unusual?	Anomaly detection
4	Which districts will have high risk next month?	Prediction

The main difficulty: A bad number and a bad service can look the same. If a district has low activity, there are two possible causes:

- The service in the district is weak.
- The data for the district is incorrect.

These causes need different actions. Thus, the project first separates data problems from real problems.

2. Domain basics

Term	Meaning
Aadhaar	India's 12-digit identity number. It uses demographic and biometric data
UIDAI	The authority that operates Aadhaar

Term	Meaning
Enrolment	A person receives Aadhaar for the first time
Demographic update	A person changes data such as name, address or date of birth
Biometric update	A person updates fingerprints, iris or face data
Mandatory biometric update	Children update biometrics at set ages, near 5 and 15 years
Saturation	Aadhaar holders as a percentage of the population
Pincode	A six-digit postal area code
District	An administrative area in a state

Why child biometric updates are important: Fingerprints and iris data change as a child grows. If a child does not update, identity checks can fail. This can affect access to services.

Why new districts are important: States sometimes divide a district into two districts. The new district receives a new name. Different data systems can start to use the new name at different times. Chapter 11 shows the effect of this.

Part B: Understanding the data

3. What one row means

The problem: Many analysis errors occur because the analyst does not know what one row represents.

The concept: The *grain* of a dataset is the meaning of one row. Always identify the grain first.

In this project, one row is not one person. One row is a count for one date and one pincode.

date	state	district	pincode	bio_age_5_17	bio_age_17_
05-09-2025	Bihar	Patna	800001	12	9

This row is an example to show the structure. It is not from the data. It says that 21 people did biometric updates in that area on that date.

Why this matters: Two totals are possible.

Total	Calculation	Meaning
Row count	Number of rows	How data was reported
People count	Sum of count columns	Actual activity

Use the people count for activity. A row count measures the reporting format.

Datasets:

Dataset	Count columns	Rows after cleaning
Enrolment	`age_0_5`, `age_5_17`, `age_18_greater`	976,002
Demographic updates	`demo_age_5_17`, `demo_age_17_`	1,591,929

Dataset	Count columns	Rows after cleaning
Biometric updates	`bio_age_5_17`, `bio_age_17_`	1,761,911

Questions that you can now answer:

- What is the grain of the data?
- Why is a row count not a measure of activity?
- Why can one person not be tracked in this data?

4. The data audit

The problem: A trend is valid only if each period is measured in the same way.

The concept: A data audit measures the structure of the data before analysis. It does not look for insights. It looks for changes in structure.

The audit: For each month and dataset, count the rows, dates and people. Then calculate people for each row.

Month	Biometric rows	Dates	People	People for each row
July	22,169	1	9,792,552	441.7
August	0	0	0	No data
September	416,846	20	6,654,928	16.0
October	331,715	18	4,582,655	13.8
November	459,086	20	7,285,506	15.9
December	543,758	26	8,704,308	16.0

How to read the table: In July, one date holds all the data. Each row has 441.7 people. In September, the data is spread across many dates. Each row has 16 people.

Thus, July is a monthly summary. September is daily data. The two formats are not comparable by row.

Across all datasets, rows increase approximately 50 times in September. People increase by only a small amount.

Other gaps:

- August is not in any file.
- Biometric and demographic data stop on 20 September.
- All datasets have no data for 1 to 12 October.
- December enrolment has only 11 dates.

The decision: Use only September to December for trends. Add a `period_type` column to mark each row as `snapshot` or `daily`.

What can go wrong: If you compare row counts between July and September, you find a false increase. It is a change in reporting. It is not a change in demand.

Questions that you can now answer:

- How did you find the format change?

- Why did you not use March to July for trends?
- What does people for each row show?

Part C: Preparing the data

5. Cleaning the data

The problem: One real place can have many names. One name can also represent two real places.

Issue	Example	Effect
Spelling	West Bangal	Creates a false state
Old name	Orissa	Splits Odisha into two
Punctuation	Baghpat *	Creates a false district
Shared name	Aurangabad	Merges two real districts
Wrong location	Rupnagar under Chandigarh	Puts data in the wrong state

The cleaning sequence:

1. Parse the dates.
2. Standardize state names, then keep only the 36 valid states and territories.
3. Keep pincodes with six digits.
4. Correct each state with its pincode.
5. Standardize district names.
6. Remove districts with fewer than 20 rows.
7. Remove duplicate rows.

Concept: Sequence is important. Duplicate removal comes last. Two rows can have different spellings. After standardization, they become exact duplicates. If you remove duplicates first, you cannot see them.

Concept: Pincode majority vote. A pincode belongs to a specific area. For each pincode, the code finds the most frequent state. It then applies this state to all rows with that pincode.

```
pin_state = (
    pd.concat([df[['pincode', 'state']] for df in datasets.values()])
    .value_counts().reset_index(name='rows')
    .drop_duplicates('pincode')
    .set_index('pincode')['state']
)
```

Worked example: A pincode has 980 rows under Punjab and 20 rows under Chandigarh. The majority state is Punjab. The code moves the 20 rows to Punjab. This step moved approximately 101,000 rows, or 2% of the data.

Concept: Context-specific mapping. A simple rename applies everywhere. Bihar has an Aurangabad district. Maharashtra has a different Aurangabad, which is now Chhatrapati Sambhajinagar. Thus, the rename applies only in Maharashtra. The same rule applies to Bijapur in Karnataka and Chhattisgarh.

Result:

Step	Rows after step	Rows removed
Raw data	4,938,837	
Valid dates and states	4,938,801	36
Valid pincodes	4,938,801	0
Valid districts	4,938,491	310
Duplicates removed	4,329,842	608,649

State and district pairs decrease from 1,077 to 907.

What can go wrong:

- A majority vote can be incorrect at a state boundary.
- The data is aggregated. Two identical rows can be real, but this is unlikely.
- 907 is more than the official number of districts. Some variants remain.

Questions that you can now answer:

- Why did you remove duplicates last?
- Why did you use pincodes to correct states?
- Why is a simple rename dangerous?

6. Joining the datasets: missing is not zero

The problem: The analysis needs enrolment, biometric and demographic data in one table.

The concept: A join connects tables with a shared key. Here, the key is state and district.

Join	Keeps	Risk
Left	All rows from the left table	Drops rows that exist only on the right
Inner	Only rows in both tables	Drops many rows
Outer	All rows from all tables	Creates missing values

This project uses an outer join and keeps the missing values.

Why missing is not zero:

Value	Meaning
0	The data was reported, and no activity occurred
Missing	The data does not tell us

Worked example: A district has 100 enrolments and no biometric row.

Method	Biometric value	Conclusion
Fill with 0	0	The district has no biometric service
Keep missing	Missing	The biometric activity is not known

The first conclusion can cause a wrong decision. The second conclusion is correct.

Effect in this project: No district with sufficient data has zero child biometric updates. 36 districts have too few updates to judge. One district has no biometric rows.

Questions that you can now answer:

- Why did you use an outer join?
- What is the difference between zero and missing?

Part D: Metrics

7. How to calculate a correct ratio

The problem: A state has many districts. How do you calculate the state ratio?

The concept: Three methods give different answers.

Worked example:

District	Biometric	Child updates	Ratio
A	900	1,000	0.90
B	10	100	0.10
Total	910	1,100	

Method	Calculation	Result	Problem
Sum of ratios	0.90 + 0.10	1.00	A share cannot be calculated this way
Mean of ratios	(0.90 + 0.10) / 2	0.50	Small District B has the same weight as large District A
Ratio of sums	910 / 1,100	0.827	Correct. Each update has equal weight

The rule: Sum the counts first. Then divide.

Why the sum of ratios is dangerous: The result increases with the number of districts. Uttar Pradesh has many districts. If you add monthly district ratios, its score is 67. A share cannot be 67. Across states, this score has a 0.98 correlation with the number of districts. Thus, this score measures the number of districts.

Correct result:

State	Child biometric share
Manipur	74.1%
West Bengal	74.9%
Bihar	75.1%
Telangana	78.7%
Meghalaya	78.9%
Uttar Pradesh	84.2% (16th lowest of 36)

Sanity check: A share must be between 0 and 1. The notebook has an `assert` statement to make sure of this.

Questions that you can now answer:

- What is the difference between a mean of ratios and a ratio of sums?
- How can you detect this error?

8. Definition of each metric

The problem: A metric name must describe what the formula measures.

Metric	Formula	What it measures
Child biometric share	`bio_5_17 / (bio_5_17 + demo_5_17)`	Biometric part of child updates
Demographic share	Demographic updates / all updates	Mix of update types
Adult enrolment share	`age_18_greater` / total enrolment	Adult part of new enrolment
Update intensity	Updates / enrolments	Updates relative to enrolment
Updates for each 1,000 people	Updates / population × 1,000	Load for each resident

Numerator and denominator: A rate needs the correct denominator.

Child biometric share uses child updates as its denominator. It does not use the children who must update. Thus, it is not a compliance rate.

Value	Child biometric share	Compliance rate
Numerator	Child biometric updates	Completed mandatory updates
Denominator	Child biometric plus demographic updates	Children who must update
Available in this data?	Yes	No

Example: A district has 80 child biometric updates and 20 child demographic updates. The share is 80%. But 1,000 children can be due. Then true completion is only 8%. The share cannot show this.

Questions that you can now answer:

- Why is this metric not compliance?
- What data do you need for true compliance?

9. Small numbers and uncertainty

The problem: A percentage from a small count is not stable.

The concept: The standard error of a proportion shows the expected variation.

```
standard error = square root of ( p × (1 - p) / n )
95% range      = p ± 1.96 × standard error
```

Worked example: The true share is 0.80.

Child updates (n)	Standard error	Approximate 95% range
5	0.179	0.45 to 1.00
30	0.073	0.66 to 0.94
900	0.013	0.77 to 0.83

With 5 updates, the share is almost meaningless. With 30 updates, it is usable. With 900 updates, it is precise.

The decision: Calculate a share only if a district has 30 or more child updates. The value 30 is a judgment. A different value changes the number of districts, but it does not change the logic.

```
(bio / child_updates).where(child_updates >= MIN_CHILD_UPDATES)
```

Questions that you can now answer:

- Why did you use a minimum of 30?
- Why do small districts appear more unusual?

10. Comparing fairly: per day and per person

The problem: Two values are comparable only if they have the same base.

Concept: Per reporting day. Months have different numbers of dates in the data. Thus, monthly totals are not comparable.

Month	Biometric dates	Biometric people per day
September	20	325,572
October	18	235,138
November	20	338,448
December	26	315,569

Concept: Per person. Large states have large totals. A per-person rate shows the load.

Measure	Result
Top five states	51.6% of all updates
Manipur	102 updates for each 1,000 people
State median	31 updates for each 1,000 people
Manipur compared with median	3.3 times

Why both measures are necessary: Volume shows where to put total capacity. Per-person load shows which system has high stress.

Peak analysis: September is the busiest month, per reporting day, in 22 states. November is the busiest in 4 states. In Maharashtra, November is the busiest month. Thane and Pune approximately double from October to November. In Pune, demographic updates increase from 33,436 to 89,260.

Concept: Correlation. Is high adult enrolment related to low saturation?

Spearman correlation compares ranks. It is less sensitive to outliers than Pearson correlation. Its value is between -1 and +1.

- Result: -0.22
- P-value: 0.20
- States: 36

A p-value is the probability of a result this strong if no real relation exists. At 0.20, the data does not show a reliable relation across states.

Meghalaya is a specific case. It has 37% adult enrolment and 84% saturation. Nagaland has the lowest saturation, at 62%.

Questions that you can now answer:

- Why did you normalize by reporting day?
- Why do you need volume and per-person load?
- What does a p-value of 0.20 mean?

Part E: The pincode test

11. Finding the true cause of low districts

The problem: 26 districts have a child biometric share below 50%. Is the service weak, or are the records incorrect?

Observation: Many of these districts are new. Examples are Didwana-Kuchaman, Kotputli-Behror and Khairthal-Tijara.

Hypothesis: The demographic data uses the new district name. The biometric data still uses the old parent name.

Why a pincode is a good test: A pincode is more stable than a district name. A new district does not change the pincode. Thus, the pincode can connect records with different district names.

Method:

1. Select a low district.
2. Find its pincodes in the demographic data.
3. Find all biometric rows for those pincodes.
4. Group these rows by district name.
5. Calculate the share for all names together.

```
share by pincode = biometric for pincodes /
                   (biometric for pincodes + demographic for pincodes)
```

Worked example:

Item	Value
District	Didwana-Kuchaman
Child biometric under own name	1

Item	Value
Child biometric under Nagaur	26,290
Share as reported	0.018
Share by pincode	0.912

Concept: Like-for-like comparison. Both sides of a ratio must cover the same population. The formula above uses all district names for the biometric count and the demographic count. If the demographic count uses only the new name, the share is too high.

Results:

Measure	Value
Districts tested	26
Most records under a different name	24
Median share as reported	0.27
Median share by pincode	0.85
Districts at 70% or more	20

Other parent names include Alwar, Bharatpur, Barpeta, Karimganj and Ramanagar.

Two districts remain. Shamator and Kamle have no records under another name. They can have real low activity, or too little data.

Effect on the decision:

If you trust the raw share	If you use the pincode test
Send field teams to 26 districts	Correct records for 24 districts
High cost	Low cost
No improvement if the cause is names	Direct action on the cause

What can go wrong:

- A pincode can cross a district boundary.
- The test uses all data for a pincode, so 0.85 is an estimate.
- The test shows the pattern. It does not prove the reason for the different names.

How to confirm: Compare the records with the official district master list. Ask UIDAI how each dataset assigns district names.

Questions that you can now answer:

- Why is a pincode better than a district name?
- Why must both sides of the ratio use the same pincodes?
- Is this proof?
- What are the two unexplained districts?

Part F: Anomaly detection

12. Foundations

The problem: UIDAI must review unusual districts, but no label identifies a true anomaly.

The concept: Unsupervised learning finds structure without labels. It answers the question "Which points are unusual?" It does not answer the question "Which points are bad?"

Features: A feature is a numeric description of a district.

Feature	Shows
Child biometric share	Child update mix
Demographic share	General update mix
Adult enrolment share	Late enrolment
Log update intensity	Updates relative to enrolment
Peak-day share	Concentration on one date
Weekend share	Activity on weekends

Why no size feature: If size is a feature, large districts appear unusual because they are large.

Why a logarithm: Update intensity is skewed. A few districts have very large values. A logarithm decreases this skew.

Concept: Scaling. Features have different ranges. `RobustScaler` uses this formula:

```
scaled value = (value - median) / interquartile range
```

The median and interquartile range are not sensitive to extreme values. Thus, outliers do not change the scale.

Worked example: Child biometric share has a median of 0.843 and an interquartile range of 0.098 (0.886 - 0.788). For a district at 0.071:

```
(0.071 - 0.843) / 0.098 = -7.9
```

The district is 7.9 interquartile ranges below the median.

13. Z-score

The concept: A z-score shows the distance from the mean.

```
z = (value - mean) / standard deviation
```

The code flags a value if the absolute z-score is 3 or more.

Strength: It is simple and easy to explain.

Limit: It checks one feature at a time. A district can be normal on each feature, but unusual in the combination.

Result: 59 districts are flagged.

14. Isolation Forest

The concept: Unusual points are easy to separate from the other points.

How it works:

1. Select a random sample of the data.
2. Select a random feature.
3. Select a random split value.
4. Divide the points into two groups.
5. Continue until each point is alone.
6. Repeat to make many trees.
7. Count the average number of splits to isolate each point.

The key idea: A normal point is in a dense area, so it needs many splits. An unusual point is alone, so it needs few splits.

```
Normal point:    many splits  ->  long path   ->  low anomaly score
Unusual point:   few splits   ->  short path  ->  high anomaly score
```

Worked example: Consider these values for one feature: 0.80, 0.81, 0.82, 0.83, 0.10. A random split between 0.10 and 0.80 isolates 0.10 at once. This split has a high probability, because the gap is large. The value 0.82 needs more splits.

Settings in this project:

Setting	Value	Meaning
`n_estimators`	300	Number of trees
`contamination`	0.05	Flag 5% of districts
`random_state`	7	Gives a repeatable result

Important: `contamination` does not discover the true anomaly rate. It sets the score cut-off. Here, it agrees with a review capacity of approximately 43 districts.

15. Local Outlier Factor

The problem: Isolation Forest finds points that are far from the data. Some points are unusual only in relation to their neighbours.

The concept: Compare the local density of a point with the density of its neighbours.

How it works:

1. Find the 20 nearest neighbours of a point.
2. Calculate how dense the area is near the point.

3. Calculate how dense the area is near its neighbours.
4. Divide the neighbour density by the point density.

```
LOF near 1        ->  same density as neighbours  ->  normal
LOF much above 1  ->  less dense than neighbours  ->  outlier
```

Worked example: Consider a tight group of districts. Their neighbours are very close. One district is near this group, but not in it. Its global values look normal. However, its nearest neighbours are in a much denser area. Thus, its LOF value is high.

Isolation Forest compared with LOF:

Item	Isolation Forest	LOF
Asks	Is the point easy to separate?	Is the point less dense than its neighbours?
View	Mostly global	Local
Speed	Fast	Slower on large data
Sensitive to	Random seed	Number of neighbours

16. Validating results without labels

The problem: No answer key exists. How can you trust the flags?

Check 1: Agreement. Two methods flagged 43 of 842 districts each. If they were independent, how many would overlap?

```
expected overlap = 43 × 43 / 842 = 2.2
```

The actual overlap is 25. This is approximately 11 times more than chance. The rank correlation of the two scores is 0.76.

Check 2: Stability. Isolation Forest uses random values. The code ran it with 20 different seeds. On average, a flagged district stayed flagged in 95% of runs.

Check 3: Bias. The anomaly score has a correlation of -0.41 with district size. Thus, small districts receive more flags. Chapter 9 gives the reason: their percentages are less stable.

Check 4: Explanation. For each lead, the code finds the feature that is furthest from normal. A person can then review the reason.

Results:

- 43 districts are flagged by two or more of the three methods.
- 13 are the name problem from Chapter 11.
- Meghalaya has the largest group, with 11 districts. Adult enrolment is the main reason.

Key lesson: An anomaly is a lead. It is not a verdict. Here, the models found real unusual districts. Many are unusual because of a data problem.

17. Unusual dates

The problem: Which dates are unusual? Sundays are always low. A simple average gives false flags.

The method:

1. Find the median for each weekday.
2. Calculate the log ratio of each date to this median.
3. Calculate a robust z-score.
4. Flag dates with a score of 3 or more.

```
log ratio = log(updates on date / weekday median)
robust z  = (log ratio - median) / (1.4826 × MAD)
```

MAD is the median absolute deviation. The value 1.4826 makes MAD comparable to a standard deviation for normal data.

Worked example: On 22 October, updates are 9,601. The Wednesday median is 604,457.

```
log(9,601 / 604,457) = -4.14
```

This is a very large decrease. The robust z-score is -20.8.

Validation: The method flagged 20 to 23 October, near Diwali. The method did not use a holiday calendar. Thus, the method finds real events. It flagged 10 of 82 dates. Some flagged dates are next to data gaps, so they need a review.

Why not Prophet or LSTM: These methods need long continuous series. This data has approximately 90 dates with gaps. A simple baseline is more reliable.

Questions that you can now answer:

- How does Isolation Forest work?
- What is the difference between Isolation Forest and LOF?
- What does contamination do?
- How do you validate without labels?
- How did you calculate 2.2?
- Why are small districts flagged more?
- Why compare a date with its weekday?

Part G: Prediction

18. Defining the prediction problem

The problem: UIDAI wants to act before a problem becomes worse.

The target: Will this district be in the bottom 25% for child biometric share next month?

Concept: Supervised learning. Each row has features and a known answer. The model learns the relation between them.

Row	Features from	Target from
Training 1	September	October
Training 2	October	November
Test	November	December

Concept: Data leakage. Leakage occurs when training uses data that is not available at prediction time. The model then appears better than it is.

Why a time-based split: A random split mixes months. The model can learn from the same future period that it is tested on. A time-based split copies real use.

Sizes: The training data has approximately 1,650 rows. The test data has approximately 830 rows. The high-risk rate is approximately 24.5%.

19. The baseline: persistence

The concept: Persistence assumes that the current pattern continues. Districts with a low share now will have a low share next month.

Why a baseline is necessary: A model score has no meaning alone. A score of 0.89 can be good or bad. You must compare it with a simple method.

```
scores = {'Persistence': -X_test['child_biometric_share'].to_numpy()}
```

The minus sign makes a low share give a high risk score.

20. Logistic Regression

The concept: Logistic Regression calculates a probability from a weighted sum of features.

```
score = b0 + b1 × feature1 + b2 × feature2 + ...
probability = 1 / (1 + e^(-score))
```

The sigmoid function changes any score to a value between 0 and 1.

How to read a coefficient: A coefficient changes the log odds. Features are standardized, so one unit is one standard deviation.

Worked example: The child biometric share coefficient is -1.48.

```
e^(-1.48) = 0.23
```

If the share increases by one standard deviation, the odds of high risk are multiplied by 0.23. Thus, the odds decrease by 77%.

Feature	Coefficient	Meaning
Child biometric share	-1.48	A high share decreases risk

Feature	Coefficient	Meaning
Log child updates	-0.76	More child updates decrease risk
Log updates	+0.47	More total updates increase risk
Demographic share	+0.42	More demographic updates increase risk

Caution: Correlated features share their effect. A coefficient shows an association. It does not show a cause.

Scaling: The model uses `StandardScaler`. The regularization penalty treats all coefficients in the same way. Thus, features must have the same scale.

21. Random Forest

The concept: A Random Forest averages many decision trees.

How it works:

1. Make a random sample of the rows for each tree.
2. At each split, test a random group of features.
3. Grow each tree.
4. Average the predictions of all trees.

Why it works: One tree can overfit. Many different trees make different errors. The average cancels part of these errors.

Settings:

Setting	Value	Meaning
`n_estimators`	400	Number of trees
`min_samples_leaf`	5	Each leaf needs 5 or more rows, which decreases overfitting

22. Evaluation metrics

The confusion matrix:

	Predicted high risk	Predicted low risk
Actual high risk	True positive (TP)	False negative (FN)
Actual low risk	False positive (FP)	True negative (TN)

```
precision = TP / (TP + FP)   Of the flagged, how many are correct?
recall    = TP / (TP + FN)   Of the true cases, how many did we find?
```

ROC AUC: Select one high-risk district and one low-risk district at random. ROC AUC is the probability that the model gives the high-risk district a higher score.

ROC AUC	Meaning
0.5	Random

ROC AUC	Meaning
0.9	Strong ranking
1.0	Perfect ranking

ROC AUC does not need a threshold. It measures rank quality.

PR AUC: This metric focuses on the positive class. A random model scores approximately the base rate, here 0.245.

Precision in top 10%: A team can review only a short list.

```
Test districts:        832
Top 10%:               83 districts
Precision 0.94:        approximately 78 are correct
```

Without a model, approximately 20 of 83 are correct.

Recall in top 25%:

```
High-risk districts:   approximately 204
Top 25%:               208 districts
Recall 0.735:          approximately 150 are found
```

Results:

Method	ROC AUC	PR AUC	Precision top 10%	Recall top 25%
Persistence	0.901	0.812	0.94	0.735
Random Forest	0.899	0.817	0.94	0.735
Logistic Regression	0.893	0.813	0.94	0.755

23. Is a difference real?

The problem: One test set gives one number. A small difference can occur by chance.

The concept: A bootstrap estimates this variation.

1. Sample test rows with replacement.
2. Calculate both scores.
3. Calculate the difference.
4. Repeat 2,000 times.
5. Find the middle 95% of differences.

Result: Logistic Regression minus persistence is -0.021 to 0.004. The interval includes zero. Thus, the data does not show a real difference.

24. Why machine learning did not win

Reason 1: The signal is already in one feature. The current share strongly predicts the next share. Other features add little.

Reason 2: Little history. The training data covers only two month pairs.

Reason 3: Stable patterns. Low districts stay low. A simple rule can capture this.

Test for the name problem: Name-problem districts always stay low. Thus, they can make all methods look good. The code removes 18 such districts from the test. ROC AUC stays near 0.89. Thus, the result does not depend on them.

The decision:

Option	Performance	Cost	Decision
Persistence	0.901	Very low	Use now
Random Forest	0.899	Higher	Test again later

A result where machine learning does not win is still a useful result. It prevents the cost of a model that adds no value.

Limits of the scores: Use the scores to rank districts. Do not read them as true probabilities, because the data is too small for calibration.

Questions that you can now answer:

- What is data leakage?
- Why did you use a time-based split?
- How do you read a coefficient?
- What is the difference between ROC AUC and precision?
- What does a bootstrap do?
- Why use the simple rule?

Part H: From analysis to decision

25. Recommendations and their logic

Recommendation	Evidence	Reason for sequence
1. Correct the district records	24 of 26 low districts	Low cost, and it improves all later analysis
2. Review child updates in Manipur, West Bengal and Bihar	Lowest shares	West Bengal and Bihar are large
3. Make enrolment plans for Meghalaya and Nagaland	37% adult share, 62% saturation	Local problems need local plans
4. Plan for high demand in Maharashtra	Thane and Pune doubled	Demand can change quickly
5. Publish a data dictionary	Two formats and gaps	Prevents false conclusions

Recommendation	Evidence	Reason for sequence
6. Use a simple watchlist	Persistence equals ML	Low cost

26. What the project cannot prove

Limit	Effect	Data needed
No count of children due	Cannot measure compliance	Birth-cohort data
Only four daily months	Weak trends	A longer period
One test month	Weak model evidence	More test months
Aggregated data	Cannot track people	Not necessary for this task
District names incomplete	District results need care	Official district master
No external event data	Cannot prove cause	Campaign and deadline records

Correlation and cause: The data shows what happened. It does not show why. For example, the November increase can have several causes.

27. How to put this into production

```
New data -> Quality checks -> Cleaning -> Metrics -> Watchlist -> Dashboard
                 |
                 +-> Alert if rows, dates or names change
```

Quality checks to include:

- Number of dates in each month
- Rows and people for each period
- New district names
- Shares outside 0 to 1
- Pincodes with more than one state

Monitoring: Measure the watchlist results each month. Train the model again only if it performs better than persistence.

Part I: Handling questions

28. How to reason through a question

Use these steps if a question is new:

1. **State the objective.** What decision does this affect?
2. **State the grain.** What does one row mean?
3. **State the assumption.** What must be true?

4. **Give a simple test.** How can you check it?
5. **State the limit.** What can the data not prove?

Example question: "How do you know that a duplicate row is really a duplicate?"

> The data is aggregated, so one row is a count for a date and pincode. An exact match on all columns is probably a repeat. However, it is not certain. I would test this by comparing the totals with a UIDAI summary. I would also ask the data owner.

If you do not know: Say so. Then say how you will find the answer.

29. Challenge questions

Is 30 the correct minimum? It is a judgment. Chapter 9 shows that a range of ±14 points is acceptable for a review. Test other values, and report if the result changes.

Why bottom 25% as the target? It gives a review list of a practical size. It is a relative target. Thus, some districts will always be high risk. An absolute target needs a true compliance rate.

Is a relative target a problem? Yes, it can be. If all districts improve, 25% are still flagged. Use a fixed threshold in production.

Why not use cross-validation? Standard cross-validation mixes months. A time-series split is correct, but the data has only three month pairs.

Does the model suffer from regression to the mean? Some low values are low by chance, and they will increase next month. This limits the value of persistence for very small districts. The minimum count decreases this effect.

Can the 5% flag rate miss real problems? Yes. A low rate misses some real anomalies. A high rate gives more false alarms. The review capacity decides the balance.

What if the district names are corrected? Run the analysis again. The low-district list will change. Some real problems can then become visible.

Why trust the pincode test if 0.85 is an estimate? The direction is more important than the exact value. For most districts, nearly all biometric records are under another name.

What is the largest risk in the project? The largest risk is the metric. Child biometric share does not measure the children who must update. A high share can hide a large gap.

What will you do with more time?

1. Get the official district master.
2. Get the count of children who must update.
3. Get a complete monthly series.
4. Test the watchlist on more months.

Part J: Reference

30. Notebook map

Notebook	Input	Output	Main concepts
01 Data audit and cleaning	12 CSV files	Clean Parquet files	Grain, audit, cleaning sequence
02 Corrected metrics and analysis	Clean files, saturation file	District and state metrics	Joins, ratios, pincode test
03 Anomaly detection	District metrics, daily totals	Lead list, unusual dates	Isolation Forest, LOF
04 Early-warning model	Monthly metrics	Model results, watchlist	Time split, metrics, bootstrap

31. Key numbers

Item	Value
Raw rows	4,938,837
Clean rows	4,329,842
Duplicates removed	608,649
State and district pairs	1,077 to 907
Rows moved by pincode	Approximately 101,000
Row increase in September	Approximately 50 times
Correlation of summed ratios with district count	0.98
Uttar Pradesh rank	16th lowest of 36
Low districts explained	24 of 26
Median share	0.27 to 0.85
Update mix	62% biometric, 33% demographic, 4.8% enrolment
Top five states	51.6% of updates
Manipur	102 for each 1,000 people, 3.3 times the median
Meghalaya	37% adult share, 84% saturation
Nagaland saturation	62%
Saturation correlation	-0.22, p = 0.20
Anomaly overlap	25, with 2.2 by chance
Seed stability	95%
Persistence, Random Forest, Logistic Regression	0.901, 0.899, 0.893

Item	Value
Logistic minus persistence	-0.021 to 0.004